Section 3.4

Independence, Basis and dimension.

Linear dependence and independence.

- **If at least one vector in a set can be written as a linear combination of the others, we say that the set is linearly dependent.**
- **If no vectors from the set can be written as linear combinations of the others, we say the set is linearly independent.**

$R = \{(1,0,1), (1,0,-1)\}$ **is linearly independent.**

$S = \{(1,0,1), (1,0,-1), (2,02)\}$ **is linearly dependent.**

$$(2,0,2) = 2(1,0,1)$$

$T = \{(1,0,1), (1,0,-1), (3,0,1)\}$ **is linearly dependent.**

$$(3,0,1) = 2(1,0,1) + (1,0,-1)$$

Formal definition of linear independence and linear dependence.

DEFINITION The sequence of vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ is ***linearly independent*** if the only combination that gives the zero vector is $0\boldsymbol{v}_1 + 0\boldsymbol{v}_2 + \cdots + 0\boldsymbol{v}_n$.

Linear independence

$$x_1\boldsymbol{v}_1 + x_2\boldsymbol{v}_2 + \cdots + x_n\boldsymbol{v}_n = \mathbf{0} \quad \text{only happens when all } x\text{'s are zero.} \tag{1}$$

If a combination gives $\mathbf{0}$, when the x's are not all zero, the vectors are *dependent*.

Linear dependence and independence.

- **As the rows and columns of a matrix can be considered set of vectors, we can discuss their linear dependence/independence.**

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 2 & 2 & 4 \\ 0 & 0 & 3 & 3 \end{bmatrix}$$

- **The row and columns of A are both linearly dependent.**
- **The row and columns of B are both linearly independent.**
- **The rows of C are linearly independent, but the columns of C are linearly dependent.**

Thm: The columns of A are linearly independent when the only solution to Ax = 0 is x=0.

- Example 1: Find the solution of Ax = 0 for A= $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$.

Answer: A= $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ by elimination R= $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

The only solution is x=0. Hence the columns are linearly independent.

Thm: The columns of A are linearly independent when the only solution to Ax = 0 is x=0.

- Example 2: Find the solution of Ax = 0 for A= $\begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 5 \\ 1 & 0 & 3 \end{bmatrix}$.

Answer: A= $\begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 5 \\ 1 & 0 & 3 \end{bmatrix}$ by elimination R= $\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$.

There is a column (3rd column) with free variable. Not linearly independent because (-3, 2, 1) is a solution.

$$Ax = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 5 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} \quad \text{is} \quad -3 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 3 \\ 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Dimension of a linear span.

- **The linear span of a set of vectors is the set of all possible linear combinations of those vectors; we call the set of vectors we begin with a spanning set.**
- **If a spanning set is linearly dependent, we can remove any linearly dependent vectors without changing the span because these dependent vectors can be "reconstructed" through linear combinations:**

$$R = \{(1,0,1), (1,0,-1)\}$$
$$S = \{(1,0,1), (1,0,-1), (2,02)\}$$
$$T = \{(1,0,1), (1,0,-1), (3,0,1)\}$$

$$U = span(R) = span(S) = span(T) = \{\vec{x} : \vec{x} = s(1,0,1) + t(1,0-1), \forall s,t \in \mathbb{R}\}$$

- **The dimension of a linear span is the number of *linearly independent* vectors in a spanning set or the minimum number of vectors needed to generate the span:**

$$\dim(U) = \dim(span(R)) = 2$$

Geometric meaning of dimension of a vector space

Definition: The dimension of a vector space is the number of *linearly independent* vectors in a vector space or the minimum number of vectors needed to generate the space (Similar to the definition for the dimension of linear span).

Fact 1: # vectors > the dimension, some vectors must be linearly dependent.

Fact 2: # vectors < the dimension, it can not be a spanning set. For example, a plane can not be generated by only one vector.

***Geometric interpretation:* The dimension defines its geometry:**

- **A vector space of dimension 0 is a point (no generating vectors → origin only).**
- **A vector space of dimension 1 is a line (generated by one vector).**
- **A vector space of dimension 2 is a plane (generated by two vectors).**

Basis of a vector space.

Definition: A *basis* for a vector space is a sequence of vectors that have two properties: (1) basis vectors are linearly independent and (2) the basis vectors spans the space.

- **A basis for a vector space:**
  - **Must be linearly independent → no more vectors than necessary.**
  - **Must be a spanning set → no fewer vectors than necessary.**
  - **Is not unique → any linearly independent spanning set is a *basis.***
    - **Any single vector on a line (through the origin) is a basis for that line.**
- **The dimension of a vector space:**
  - **Is the number of vectors in any basis.**
  - **Defines the geometry of the space.**
  - **Is a single, constant number.**

Facts:

- 1. There is only one way to write v as a linear combination of the basis vectors.

Reason: Suppose $\boldsymbol{v} = a_1\boldsymbol{v}_1 + \cdots + a_n\boldsymbol{v}_n$ and also $\boldsymbol{v} = b_1\boldsymbol{v}_1 + \cdots + b_n\boldsymbol{v}_n$. By subtraction $(a_1 - b_1)\boldsymbol{v}_1 + \cdots + (a_n - b_n)\boldsymbol{v}_n$ is the zero vector. From the independence of the $\boldsymbol{v}$'s, each $a_i - b_i = 0$. Hence $a_i = b_i$, and there are not two ways to produce $\boldsymbol{v}$.

Basis and dimension.

Questions:

- **Given a set of vectors that span a vector space, how do we find the dimension and a basis?**
- **Given a set of vectors, how do we know if they are linearly independent, or not?**
- **If not, how do we know which vectors are linearly independent, and which can be removed without changing the span?**

Dimension of matrix spaces

- The vector space M contains all 2 by 2 matrices. What is the dimension? What is a basis?
- Answer: Its dimension is 4.

One basis is $A_1, A_2, A_3, A_4 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

Questions: dimensions of matrix space

- What is the dimension of the whole n by n matrix?
- For an n by n matrix space, what is the dimension of the subspace of the upper triangular matrices?
- For an n by n matrix space, what is the dimension of the subspace of diagonal matrices?
- For an n by n matrix space, what is the dimension of the subspace of the symmetric matrices?

The dimension of the whole n by n matrix space is n^2.

The dimension of the subspace of *upper triangular* matrices is $\frac{1}{2}n^2 + \frac{1}{2}n$.

The dimension of the subspace of *diagonal* matrices is n.

The dimension of the subspace of *symmetric* matrices is $\frac{1}{2}n^2 + \frac{1}{2}n$ (why ?).

Relationship between Linear dependence and elimination.

During elimination:

1. If there exist non-zero coefficients for a linear combination of rows to get the zero vector, they must be linearly dependent:

$$a\vec{u} + b\vec{v} + c\vec{w} = 0 \leftrightarrow \vec{w} = \frac{a}{c}\vec{u} + \frac{b}{c}\vec{v}, c \neq 0$$

2. The rows are linearly dependent if and only if zero rows appear.

3. Every matrix generated during the process of elimination has the same row space: $A \to U \to R \iff \boldsymbol{R}(A) = \boldsymbol{R}(U) = \boldsymbol{R}(R)$

Row Space: Basis and dimension.

Facts:

1. The non-zero rows of any echelon form of a matrix are a basis for its row space.

2. # of non-zero rows of any echelon form = dimension of the row space.

Procedure: The Row Space algorithm is to find a basis from a spanning set:

- Create a matrix whose rows are the spanning set.
- Carry out elimination to either U or R.
- The resulting non-zero rows are a basis.

$$T = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m\} \to A = \begin{bmatrix} \vec{v}_1^T \\ \vec{v}_2^T \\ \vdots \\ \vec{v}_r^T \\ \vec{v}_{r+1}^T \\ \vdots \\ \vec{v}_m^T \end{bmatrix} \to\to \begin{bmatrix} \vec{u}_1^T \\ \vec{u}_2^T \\ \vdots \\ \vec{u}_r^T \\ \vec{0} \\ \vdots \\ \vec{0} \end{bmatrix} = U \text{ or } R \to \begin{array}{c} \text{Basis: } S = \{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_r\} \\ \dim(span(S)) = r \\ span(T) = span(S) \end{array}$$

Example: Basis and dimension of row space

- **Example – Basis and dimension of the row space of a matrix.**

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 2 & 3 & 3 & 4 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

$\dim\big(\boldsymbol{R}(A)\big) = 3,$ Basis of $\boldsymbol{R}(A)$: $\{(1,1,1,1), (1,1,1,2), (0,1,1,1)\}$

Alternative basis of R(A): {(1,1,1,1), (0,1,1,1), (0,0,0,1)}

Example: Basis and dimension of row space

- **Example – Basis and dimension using the row space algorithm:**

$$V = span(S), S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} \right\}$$

$$S \to A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \\ 0 & 2 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & -3 \\ 0 & 2 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix} \to\to \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = R$$

$$\dim(V) = \dim\big(\boldsymbol{R}(A)\big) = 2, \qquad \text{Basis of } V\text{: } \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

The column picture of a linear system.

- $\vec{x}$ represents the coefficients needed to express $\vec{b}$ as a linear combination of the columns of A:

$$A\vec{x} = \vec{b} \rightarrow [\vec{c}_1 \quad \vec{c}_2 \quad \cdots \quad \vec{c}_n]\vec{x} = \vec{b} \rightarrow x_1\vec{c}_1 + x_2\,\vec{c}_2 + \cdots + x_n\,\vec{c}_n = \vec{b}$$

- The column space of A is the linear span of the column:.

$$\boldsymbol{C}(A) = span(\{\vec{c}_1, \vec{c}_2, \cdots, \vec{c}_n\})$$

- There will be a solution if and only if $\vec{b}$ is in the column space of A. The solution is unique only if the columns of A are linearly independent.
- The columns of A are linearly independent if and only if the null space of A is the zero vector: no free variables or every column contains a pivot.

Elimination and the column space.

- **Elimination takes linear combination of rows.**
- **The column space is not preserved during elimination:**

$$A \to U \to R \iff \boldsymbol{C}(A) \neq \boldsymbol{C}(U), \boldsymbol{C}(A) \neq \boldsymbol{C}(R)$$

- **Elimination is driven by pivots. Columns with free variables do not impact elimination because removal of a free variable column cannot introduce a new zero row (see example). Hence the existence of the solution remains unchanged**→ these columns **can be removed and the column space is preserved!**
- **Removal of a free variable column reduces the number of solutions; a unique solution exists only if there are no free variables → linearly independent columns.**

Elimination and the column space.

- **Example:** $A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & -1 & 3 \\ 1 & 2 & 3 & -2 \end{bmatrix}, \vec{b} = \begin{bmatrix} 2 \\ 10 \\ -2 \end{bmatrix}$

$$\begin{bmatrix} 1 & 2 & 1 & 0 & 2 \\ 2 & 4 & -1 & 3 & 10 \\ 1 & 2 & 3 & -2 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 & 0 & 2 \\ 0 & 0 & -3 & 3 & 6 \\ 0 & 0 & 2 & -2 & -4 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 & 0 & 2 \\ 0 & 0 & -3 & 3 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & 2 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 0 & 1 & 4 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \to \vec{x}_p = \begin{bmatrix} 4 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \vec{x}_{n1} = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \vec{x}_{n2} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 10 \\ 1 & 3 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 2 \\ 0 & -3 & 6 \\ 0 & 2 & -4 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 2 \\ 0 & -3 & 6 \\ 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \to \vec{x}_p = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$$

Note: You can remove columns with free variables and get the echelon form to find special solutions.

Column space: Basis and dimension.

- **The number of pivots in any echelon form of a matrix is the dimension of the column space.**
- **The pivot columns of the original matrix are a basis for its column space.**

The Casting-Out algorithm to find a basis from a spanning set:

- Create a matrix whose columns are the spanning set.
- Carry out elimination to either U or R.
- The pivot columns from the original matrix are a basis.

Example: Basis and dimension of column space

- **Example – Basis and dimension of the column space of a matrix.**

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 2 & 3 & 3 & 4 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

$$\dim\big(\boldsymbol{C}(A)\big) = 3, \qquad \text{Basis of } \boldsymbol{C}(A)\text{:} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \\ 4 \end{bmatrix} \right\}$$

!!!!!! You cannot use vectors in U as a basis of C(A) because C(A) is not preserved during elimination.

Example: Basis and dimension of column space

- **Example – Basis and dimension using the casting-out algorithm:**

$$V = span(S), S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} \right\}$$

$$S \to A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 1 & -1 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 0 \\ 0 & -3 & 2 \\ 0 & -3 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix} = U$$

$$\dim(V) = \dim\big(\boldsymbol{C}(A)\big) = 2, \qquad \text{Basis of } V\text{: } \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} \right\}$$

Section 3.5

Dimensions of the four subspaces.

Recall: the null space of a matrix.

- **The set of all possible solutions to the homogenous system, $A\vec{x} = \vec{0}$, is called the null space of A.**

$$\boldsymbol{N}(A) = nullsp(A) = \left\{\vec{x}: A\vec{x} = \vec{0}\right\}$$

- **The null space is a vector space:**

$$\vec{u}, \vec{v} \in \boldsymbol{N}(A) \rightarrow k\vec{u} + l\vec{v} \in \boldsymbol{N}(A), \forall k, l \in \mathbb{R}$$

- **We can describe the null space by a set of special solutions, one per free variable:**

$$\{\vec{x} : \vec{x} = s_1\vec{x}_{n1} + s_2\vec{x}_{n2} + \cdots + s_l\vec{x}_{nl}, \forall s_i \in \mathbb{R}\}$$

Null space: Basis and dimension.

- **The special solutions are created to be linearly independent:**
  - **Because a different free variable is non-zero in each special solution, we can not create one special solution from the others.**
- **The special solutions are sufficient to describe all solutions:**
  - **The general solution is the set of linear combinations of the special solutions.**
- **The special solutions form a basis for the null space!**
- **The number of free variables in an echelon form is the dimension of the null space.**

Four fundamental subspaces

Definition: The dimension of the row space and the column space of a matrix are the same which is the rank of the matrix:

$$\mathrm{r} = \mathrm{Rank}\ (A) = \dim \boldsymbol{R}(A) = \dim \mathrm{C}(\mathrm{A})$$

- **1. The row space is C (A^T), a subspace of R^n .**
- **2. The column space is C(A), a subspace of R^m .**
- **3. The null space is N (A), a subspace of R^n .**
- **4. The left null space is N(A^T), a subspace of R^m. This is our new space.**

Note: N(A) and N(A^T) have dimensions n-r and m-r to make up the full n and m.

Row space, column space and rank.

- **Forward elimination gives the dimension of the row space by counting the number of non-zero rows in a row echelon form.**
- **Forward elimination gives the dimension of the column space by counting the number of pivots in a row echelon form.**
- **A pivot is just the first non-zero entry in a row; the number of non-zero rows and the number of pivots are equal.**

Example:

- $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
- Rank (A)= dim (R(A)) = # of pivots in rows = 2
- Null (A) = # of free variables = 1
- Hence the Null (A) = n-r = 3-1=2.

Parent spaces of an m-by-n matrix.

- **Each row of an m-by-n matrix is a vector with n elements.**
- **The parent space of the row space is the n-D space.**

$$A \in \mathbb{R}^{m \times n} \rightarrow \boldsymbol{R}(A) \subseteq \mathbb{R}^n$$

- **Each column of an m-by-n matrix is a vector with m elements.**
- **The parent space of the column space is the m-D space.**

$$A \in \mathbb{R}^{m \times n} \rightarrow \boldsymbol{C}(A) \subseteq \mathbb{R}^m$$

- **Note on notation: $\subseteq$ reads as "is a subset of" or "is a subspace of"**

Relationship of null and row spaces.

- **The row and null space share the same parent space:**

$$A \in \mathbb{R}^{m\times n} \rightarrow \boldsymbol{R}(A) \subseteq \mathbb{R}^n, \boldsymbol{N}(A) \subseteq \mathbb{R}^n$$

- **The sum of the dimensions of the row and null spaces (a subspace of R^n) equals the dimension of the parent space:**

$$\dim\big(\boldsymbol{R}(A)\big) + \dim\big(\boldsymbol{N}(A)\big) = \dim(\mathbb{R}^n)\,, A \in \mathbb{R}^{m\times n}$$

- **The dimension of the row space is the rank; we give a special name to the dimension of the null space as well — the nullity:**

$$\text{nullity}(A) \equiv \dim\big(\boldsymbol{N}(A)\big)$$

$$\text{rank}(A) + \text{nullity}(A) = n\,, A \in \mathbb{R}^{m\times n}$$

The null and row spaces are linearly independent.

- For all vectors in the null space: $A\vec{x} = \vec{0}$.
- For each $\vec{r}_i \neq \vec{0}$:

$$A\vec{r}_i = \begin{bmatrix} \vdots \\ \vec{r}_i^T \\ \vdots \end{bmatrix} \vec{r}_i = \begin{bmatrix} \vdots \\ \vec{r}_i^T \vec{r}_i \\ \vdots \end{bmatrix} = \begin{bmatrix} \vdots \\ \|\vec{r}_i\|^2 \\ \vdots \end{bmatrix} \neq \vec{0}$$

- The non-zero rows cannot be in the null space $\rightarrow$ non-zero linear combinations of these rows cannot be in the null space; the only vector that is in both the row and null space is the zero vector.
- The union of a basis for the null space and a basis of the row space contains n linearly independent vectors $\rightarrow$ a basis for $\mathbb{R}^n$.

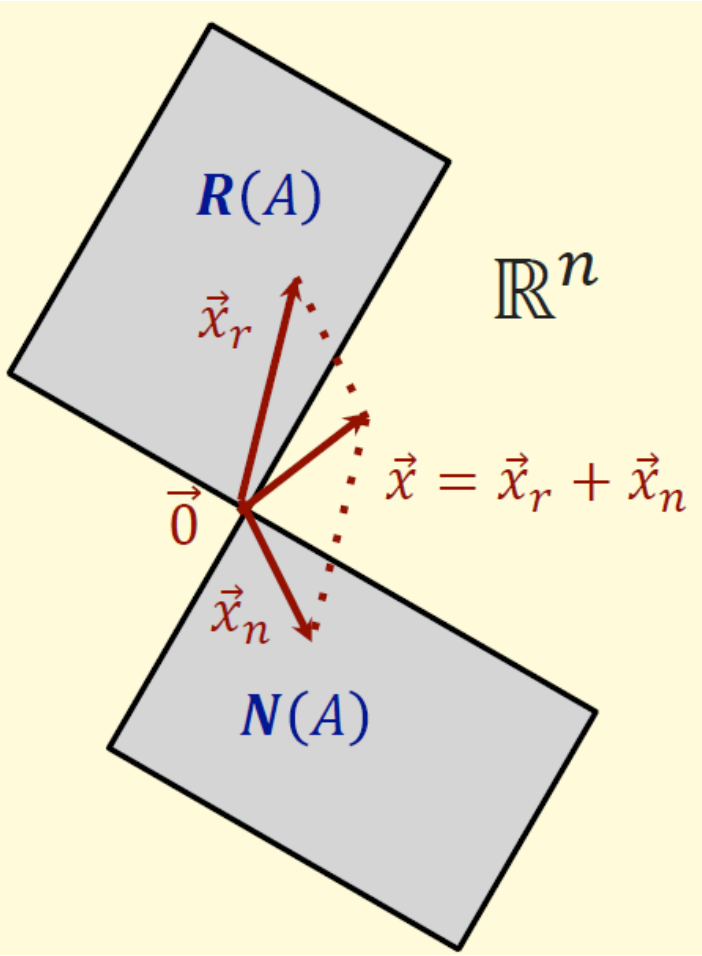
$\mathbb{R}^n$ is the direct sum of the row & null spaces.

- **A direct sum of two vector space is the space of all linear combinations of vectors from the two spaces:**

$$\boldsymbol{R}(A) \oplus \boldsymbol{N}(A) = \mathbb{R}^n$$

- **Every vector in $\mathbb{R}^n$ can be expressed as unique sum of vectors from the null space and row space:**

$$\forall \vec{x} \in \mathbb{R}^n,$$
$$\exists!\, \vec{x}_r \in \boldsymbol{R}(A), \vec{x}_n \in \boldsymbol{N}(A):$$
$$\vec{x} = \vec{x}_r + \vec{x}_n$$



Row space, null space and general solution.

- We can write the general solution for $A\vec{x} = \vec{b}$ as the sum of a particular solution and a linear combination of the special solutions to $A\vec{x} = \vec{0}$:

$$\vec{x} = \vec{x}_p + s_1\vec{x}_{n1} + s_2\vec{x}_{n2} + \cdots + s_{(n-r)}\vec{x}_{n,(n-r)}$$

- Every vector in $\mathbb{R}^n$ can be expressed as unique sum of vectors from the null space and row space:

$$\forall \vec{x} \in \mathbb{R}^n, \exists!\, \vec{x}_r \in \boldsymbol{R}(A), \vec{x}_n \in \boldsymbol{N}(A): \vec{x} = \vec{x}_r + \vec{x}_n$$

- Every solution to $A\vec{x} = \vec{b}$ has a *unique* particular solution that is in the row space.

Row and null spaces of A^T

- **The column space of any matrix is the row space of its transpose.**
- **The row space of any matrix is the column space of its transpose.**

$$\boldsymbol{C}(A) = \boldsymbol{R}\left(A^T\right), \qquad \boldsymbol{R}(A) = \boldsymbol{C}\left(A^T\right)$$

- **The row and null spaces are subspaces of $\mathbb{R}^n$:**

$$A \in \mathbb{R}^{m\times n} \rightarrow \boldsymbol{R}(A) \subseteq \mathbb{R}^n, \boldsymbol{N}(A) \subseteq \mathbb{R}^n, \qquad \boldsymbol{R}(A) \oplus \boldsymbol{N}(A) = \mathbb{R}^n$$

- **The column space is a subspace of $\mathbb{R}^m$:**

$$A \in \mathbb{R}^{m\times n} \rightarrow \boldsymbol{C}(A) \subseteq \mathbb{R}^m$$

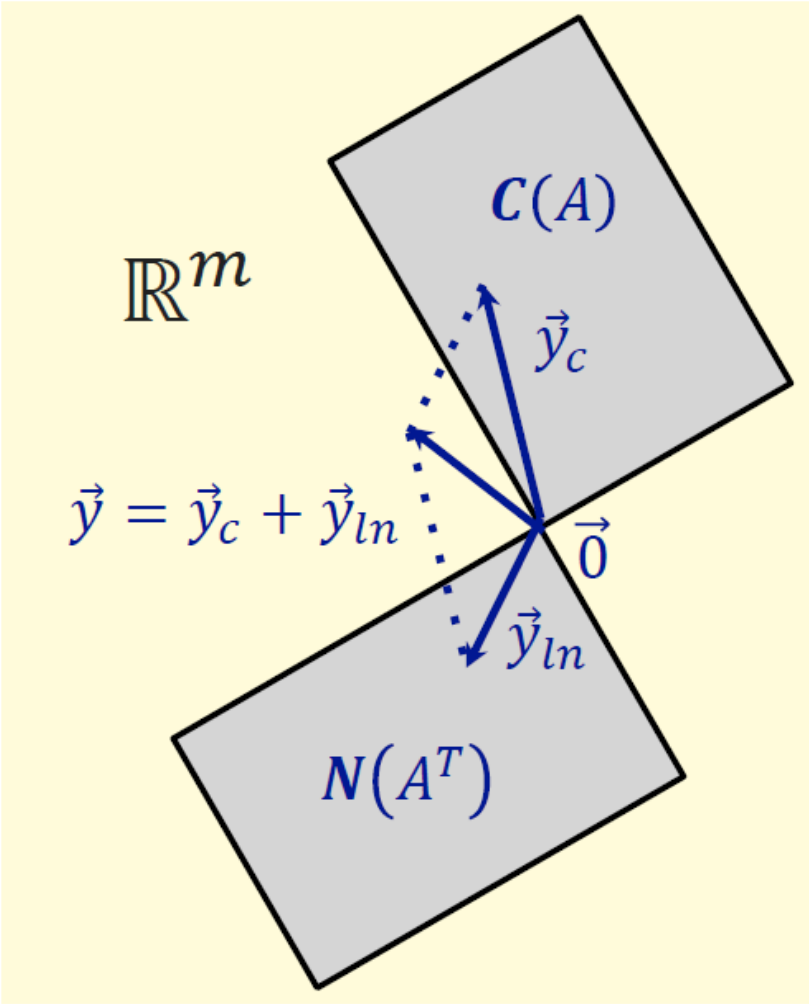
- **Define a left null space, the null space of A^T:**

$$A \in \mathbb{R}^{m\times n} \rightarrow A^T \in \mathbb{R}^{n\times m} \rightarrow \boldsymbol{N}\left(A^T\right) \subseteq \mathbb{R}^m$$

$\mathbb{R}^m$ is the direct sum of the column and left null spaces.

$$A \in \mathbb{R}^{m\times n} \rightarrow A^T \in \mathbb{R}^{n\times m}$$
$$\boldsymbol{R}(A^T), \boldsymbol{N}(A^T) \subseteq \mathbb{R}^m$$

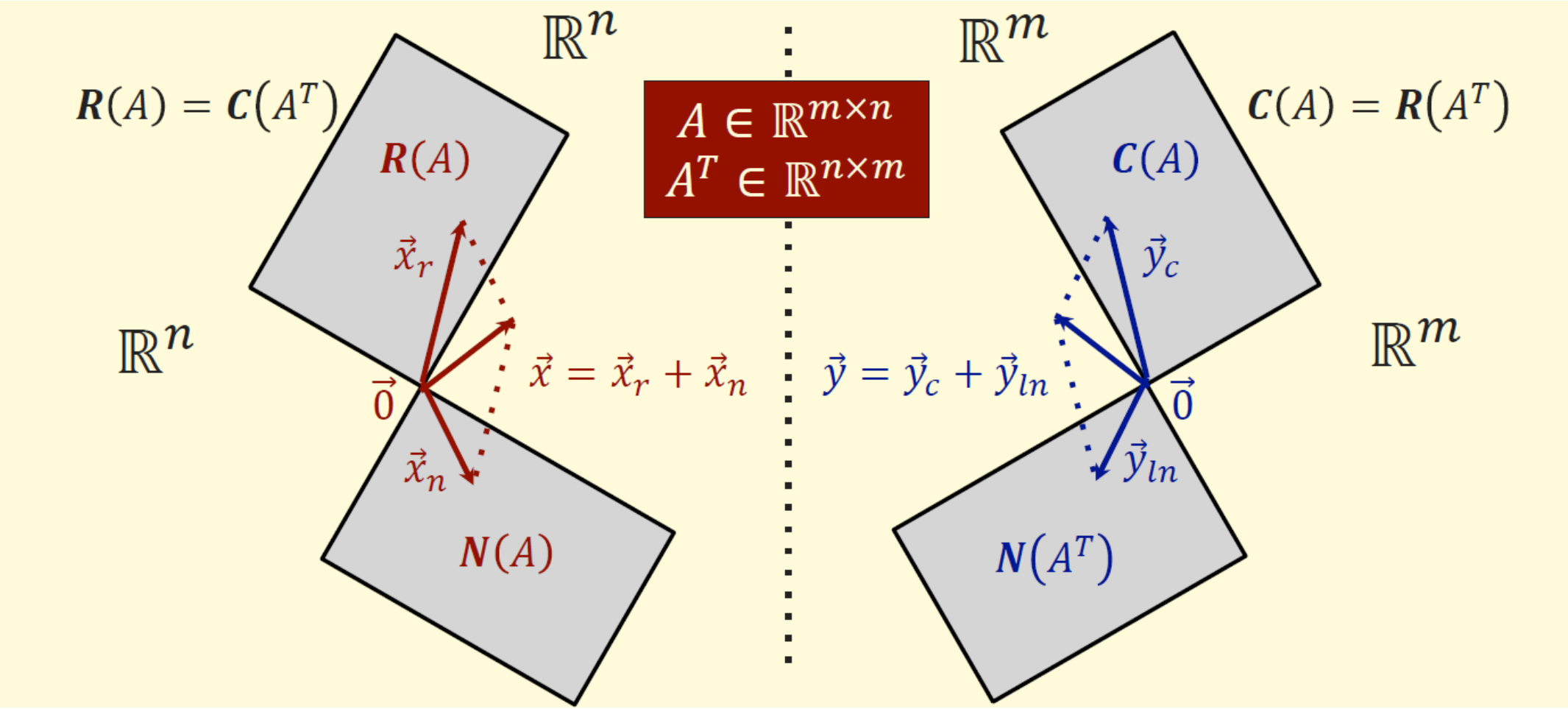
$$\boldsymbol{R}(A^T) \oplus \boldsymbol{N}(A^T) = \mathbb{R}^m$$
$$\boldsymbol{C}(A) = \boldsymbol{R}(A^T)$$



- Every vector in $\mathbb{R}^m$ can be expressed as unique sum of vectors from the column space and the left null space:

$$\forall \vec{y} \in \mathbb{R}^m,$$
$$\exists!\, \vec{y}_c \in \boldsymbol{C}(A), \vec{y}_{ln} \in \boldsymbol{N}(A^T):$$
$$\vec{y} = \vec{y}_c + \vec{y}_{ln}$$

The fundamental matrix subspaces.



One matrix, two systems.

- **For any matrix, we can define two linear systems, using either the original matrix or its transpose as the coefficient matrix.**

$$A\vec{x} = \vec{b}, A \in \mathbb{R}^{m\times n}, \vec{x} \in \mathbb{R}^n, \vec{b} \in \mathbb{R}^m$$

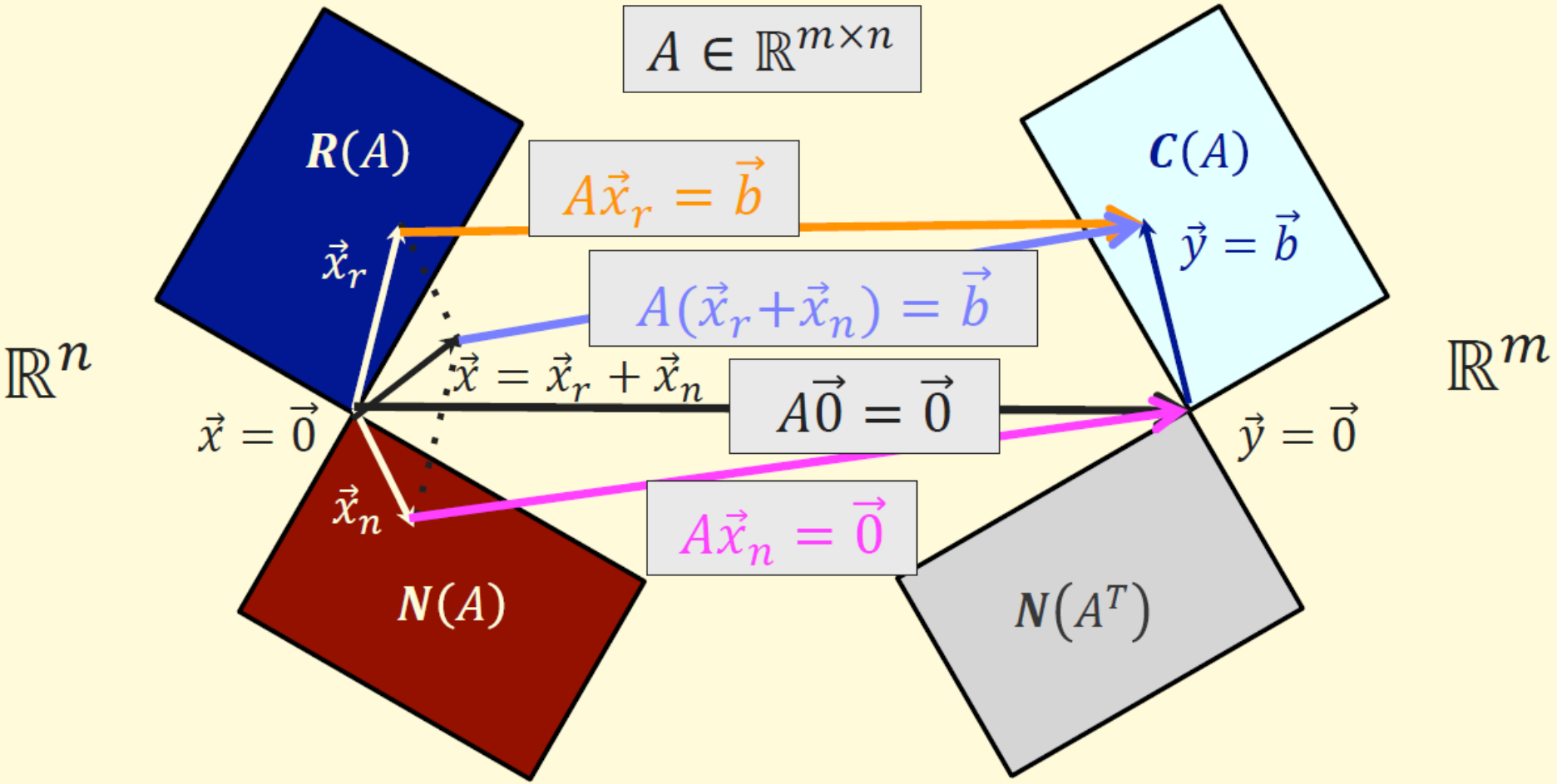
$$A^T\vec{y} = \vec{c}, A^T \in \mathbb{R}^{n\times m}, \vec{y} \in \mathbb{R}^m, \vec{c} \in \mathbb{R}^n$$

The “forward” system

$$A\vec{x} = \vec{b}, A \in \mathbb{R}^{m\times n}, \vec{x} \in \mathbb{R}^n, \vec{b} \in \mathbb{R}^m$$

- **The column space, $\boldsymbol{C}(A)$, is the set of column vectors such that the right-hand sides has a solution.**
- **The null space, $\boldsymbol{N}(A)$, is the set of solutions to $A\vec{x} = \vec{0}$.**
- **Every solution can be written as a sum of unique sum of the solution from the row space and a vector from the null space.**

The "forward" system: $A\vec{x} = \vec{b}$



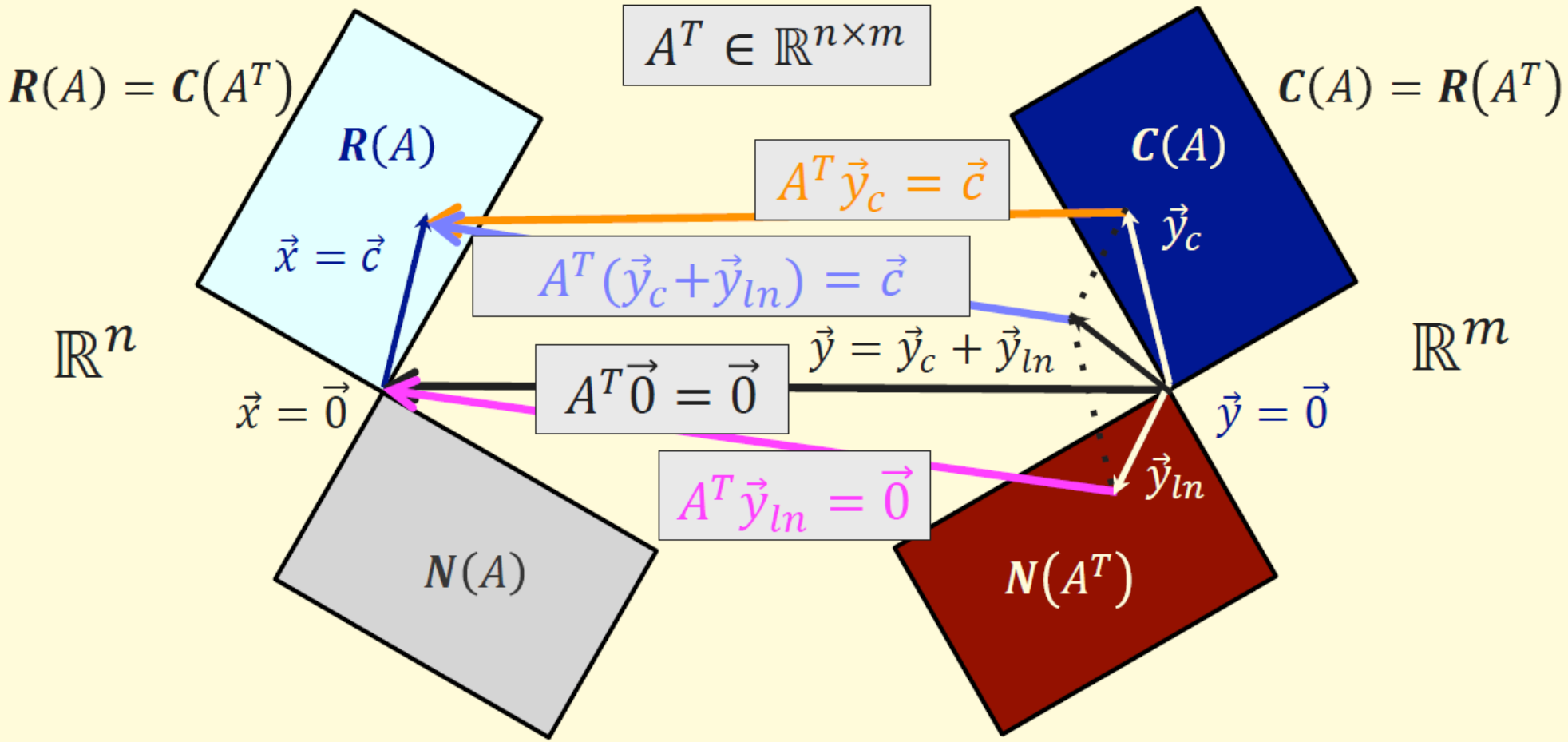
The "backward" system: $A^T\vec{y} = \vec{c}$

$$A^T\vec{y} = \vec{c}, A^T \in \mathbb{R}^{n\times m}, \vec{y} \in \mathbb{R}^m, \vec{c} \in \mathbb{R}^n$$

- The row space, $\boldsymbol{R}(A) = \boldsymbol{C}(A^T)$, is the set of right-hand sides with a solution.
- The left null space, $\boldsymbol{N}(A^T)$, is the set of solutions to $A^T\vec{y} = \vec{0}$.

- **Every solution can be written as a sum of unique sum of the solution from the column space and a vector from the left null space.**

The "backward" system: $A^T\vec{y} = \vec{c}$



Rank and the four subspaces.

Consider a coefficient matrix with *m* rows, *n* columns and a rank of *r*.

$$A\vec{x} = \vec{b},\ A \in \mathbb{R}^{m\times n}, \operatorname{rank}(A) = r$$

If $r = m = n$:

- **The matrix is full rank (and is square); nonsingular.**
- **The column space and row space are both the full parent space.**
- **The null spaces are both just the origin.**
- **There is a unique solution to every right-hand side for both $A\vec{x} = \vec{b}$,**

and $A^T\vec{y} = \vec{c}$.

Rank and the four subspaces.

Consider a coefficient matrix with *m* rows, *n* columns and a rank of *r*.

$$A\vec{x} = \vec{b},\ A \in \mathbb{R}^{m\times n},\ \mathrm{rank}(A) = r$$

If $r = m, m < n$:

- **The matrix is full row rank (and has more columns than rows).**
- **The column space is the full parent space, and the left null space is just the origin.**
- **The row space is not the full parent space, and the null space has dimension $n - r$.**
- **There are many solutions to every right-hand side for $A\vec{x} = \vec{b}$,**
- **If $A^T\vec{y} = \vec{c}$ has a solution, then it is unique, but not all $\vec{c}$ give a solution.**

Rank and the four subspaces.

Consider a coefficient matrix with *m* rows, *n* columns and a rank of *r*.

$$A\vec{x} = \vec{b}, A \in \mathbb{R}^{m\times n}, \operatorname{rank}(A) = r$$

If $r = n, m > n$:

- **The matrix is full column rank (and has more rows than columns).**
- **The row space is the full parent space, and the null space is just the origin.**
- **The column space is not the full parent space, and the left null space has dimension $m - r$.**
- **If $Ax = \vec{b}$ has a solution, then it is unique but not all $\vec{b}$ give a solution.**
- **There are many solutions to every right-hand side for $A^T\vec{y} = \vec{c}$.**

Rank and the four subspaces.

Consider a coefficient matrix with *m* rows, *n* columns and a rank of *r*.

$$A\vec{x} = \vec{b},\ A \in \mathbb{R}^{m\times n}, \operatorname{rank}(A) = r$$

If $r < m, r < n$:

- **The matrix is neither full row nor full column rank.**
- **Neither the row space nor the column space are the full parent spaces.**
- **The null space has dimension $n - r$, and the left null space has dimension $m - r$.**
- **For both $A\vec{x} = \vec{b}$ and $A^T\vec{y} = \vec{c}$, not all right-hand sides give a solution. They will give infinitely many solutions if they give a solution.**

Examples

Finding basis and dimension.

Example 1: Row space algorithm

Find a basis for the linear span of the following vectors such that the basis vectors have as many zeros as possible.

- **What is the dimension of the space?**
- **What is the parent space?**

$$V = span\left(\left\{\begin{bmatrix}1\\2\\2\\1\end{bmatrix}, \begin{bmatrix}2\\0\\1\\1\end{bmatrix}, \begin{bmatrix}3\\1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\2\\-1\end{bmatrix}\right\}\right)$$

Row space algorithm

$$\left\{\begin{bmatrix}1\\2\\2\\1\end{bmatrix},\begin{bmatrix}2\\2\\1\\1\end{bmatrix},\begin{bmatrix}3\\0\\1\\0\end{bmatrix},\begin{bmatrix}1\\-2\\2\\-1\end{bmatrix}\right\}$$

- **Create a matrix whose rows are the vectors, then carry out Gaussian elimination to reduced row echelon form**

$$A=\begin{bmatrix}1&2&2&1\\2&2&1&1\\3&0&1&0\\1&-2&2&-1\end{bmatrix}\to\begin{bmatrix}1&2&2&1\\0&-2&-3&-1\\0&-6&-5&-3\\0&-4&0&-2\end{bmatrix}\to\begin{bmatrix}1&2&2&1\\0&-2&-3&-1\\0&0&4&0\\0&0&6&0\end{bmatrix}$$

$$\to\begin{bmatrix}1&2&2&1\\0&-2&-3&-1\\0&0&4&0\\0&0&0&0\end{bmatrix}\to\begin{bmatrix}1&2&0&1\\0&-2&0&-1\\0&0&1&0\\0&0&0&0\end{bmatrix}\to\begin{bmatrix}1&0&0&0\\0&1&0&1/2\\0&0&1&0\\0&0&0&0\end{bmatrix}=R$$

Row space algorithm

$$\left\{\begin{bmatrix}1\\2\\2\\1\end{bmatrix},\begin{bmatrix}2\\2\\1\\1\end{bmatrix},\begin{bmatrix}3\\0\\1\\0\end{bmatrix},\begin{bmatrix}1\\-2\\2\\-1\end{bmatrix}\right\}$$

- **The non-zero rows are a basis; the number of nonzero rows is the dimension; the parent space is defined by the number of elements in each vector.**

$$R=\begin{bmatrix}1&0&0&0\\0&1&0&1/2\\0&0&1&0\\0&0&0&0\end{bmatrix}\rightarrow T=\left\{\begin{bmatrix}1\\0\\0\\0\end{bmatrix},\begin{bmatrix}0\\1\\0\\1/2\end{bmatrix},\begin{bmatrix}0\\0\\1\\0\end{bmatrix}\right\}$$

$$\dim\big(span(T)\big)=3 \qquad V=span(T)\subseteq\mathbb{R}^4$$

Example 2: Casting-out (column space) algorithm

Find a basis for the linear span of the following vectors such that the basis is a subset of the original set.

- What is the dimension of the space?
- What is the parent space?

$$V = span\left(\left\{\begin{bmatrix}1\\2\\1\end{bmatrix}, \begin{bmatrix}2\\4\\2\end{bmatrix}, \begin{bmatrix}1\\3\\2\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}\right\}\right)$$

Casting-out algorithm

$$\left\{\begin{bmatrix}1\\2\\1\end{bmatrix}, \begin{bmatrix}2\\4\\2\end{bmatrix}, \begin{bmatrix}1\\3\\2\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}\right\}$$

- **Create a matrix whose columns are the vectors, then carry out Gaussian elimination to echelon form:**

$$A = \begin{bmatrix}1 & 2 & 1 & 1\\2 & 4 & 3 & 1\\1 & 2 & 2 & 0\end{bmatrix} \to \begin{bmatrix}1 & 2 & 1 & 1\\0 & 0 & 1 & -1\\0 & 0 & 1 & -1\end{bmatrix} \to \begin{bmatrix}1 & 2 & 1 & 1\\0 & 0 & 1 & -1\\0 & 0 & 0 & 0\end{bmatrix} = U$$

Casting-out algorithm

$$\left\{\begin{bmatrix}1\\2\\1\end{bmatrix},\begin{bmatrix}2\\4\\2\end{bmatrix},\begin{bmatrix}1\\3\\2\end{bmatrix},\begin{bmatrix}1\\1\\0\end{bmatrix}\right\}$$

- The pivot columns of the original matrix are a basis; the number of pivots is the dimension; the parent space is defined by the number of elements in each vector.

$$A=\begin{bmatrix}1&2&1&1\\2&4&3&1\\1&2&2&0\end{bmatrix},U=\begin{bmatrix}1&2&1&1\\0&0&1&-1\\0&0&0&0\end{bmatrix}\rightarrow T=\left\{\begin{bmatrix}1\\2\\1\end{bmatrix},\begin{bmatrix}1\\3\\2\end{bmatrix}\right\}$$

$$\dim\big(span(T)\big)=2 \qquad V=span(T)\subseteq\mathbb{R}^3$$

Example 3: The four matrix subspaces.

- **Find the dimension and a basis for the four fundamental subspaces of the following matrix:**

$$A = \begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 2 & -2 & 0 & -2 & 2 \\ -1 & 1 & 1 & 2 & 0 \end{bmatrix}$$

The four matrix subspaces.

- **Begin with Gaussian elimination to reduce row echelon form:**

$$A = \begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 2 & -2 & 0 & -2 & 2 \\ -1 & 1 & 1 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 0 & 0 & -4 & -4 & -4 \\ 0 & 0 & 3 & 3 & 3 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 0 & 0 & -4 & -4 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = R$$

The four matrix subspaces.

- **The non-zero rows of an echelon form are a basis for the row space:**

$$R = \begin{bmatrix} 1 & -1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boldsymbol{R}(A) = span\left(\begin{bmatrix} 1 \\ -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}\right), \qquad \dim\big(\boldsymbol{R}(A)\big) = 2, \qquad \boldsymbol{R}(A) \subseteq \mathbb{R}^5$$

The four matrix subspaces.

- **The pivot columns of the original matrix are a basis for the column space:**

$$R = \begin{bmatrix} 1 & -1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, A = \begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 2 & -2 & 0 & -2 & 2 \\ -1 & 1 & 1 & 2 & 0 \end{bmatrix}$$

$$\boldsymbol{C}(A) = span\left(\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}\right), \qquad \dim\big(\boldsymbol{C}(A)\big) = 2, \qquad \boldsymbol{C}(A) \subseteq \mathbb{R}^3$$

The four matrix subspaces.

- **The special solutions to $R\vec{x} = \vec{0}$ are a basis for the null space:**

$$R = \begin{bmatrix} 1 & -1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \vec{x}_{n1} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \vec{x}_{n2} = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \vec{x}_{n3} = \begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\boldsymbol{N}(A) = span\left(\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}\right), \qquad \dim\big(\boldsymbol{N}(A)\big) = 3, \qquad \boldsymbol{N}(A) \subseteq \mathbb{R}^5$$

The four matrix subspaces.

- **The special solutions to $A^T\vec{y} = \vec{0}$ are a basis for the left null space:**

$$A = \begin{bmatrix} 1 & -1 & 2 & 1 & 3 \\ 2 & -2 & 0 & -2 & 2 \\ -1 & 1 & 1 & 2 & 0 \end{bmatrix}, A^T = \begin{bmatrix} 1 & 2 & -1 \\ -1 & -2 & 1 \\ 2 & 0 & 1 \\ 1 & -2 & 2 \\ 3 & 2 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & -4 & 3 \\ 0 & -4 & 3 \\ 0 & -4 & 3 \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & 2 & -1 \\ 0 & -4 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -3/4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 1/2 \\ 0 & 1 & -3/4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = R$$

The four matrix subspaces.

- **The special solutions to $A^T\vec{y} = \vec{0}$ are a basis for the null space:**

$$R = \begin{bmatrix} 1 & 0 & 1/2 \\ 0 & 1 & -3/4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \vec{x}_{ln1} = \begin{bmatrix} -1/2 \\ 3/4 \\ 1 \end{bmatrix}$$

$$\boldsymbol{N}(A^T) = span\left(\begin{bmatrix} -1/2 \\ 3/4 \\ 1 \end{bmatrix}\right), \qquad \dim\left(\boldsymbol{N}(A^T)\right) = 1, \qquad \boldsymbol{N}(A^T) \subseteq \mathbb{R}^3$$

The four matrix subspaces.

- **Double check that the dimensions all match up properly.**

$$\mathrm{Dim}\big(\boldsymbol{R}(A)\big) = 2, \qquad \boldsymbol{R}(A) \subseteq \mathbb{R}^5$$

$$\dim\ \big(\boldsymbol{N}(A)\big) = 3, \qquad \boldsymbol{N}(A) \subseteq \mathbb{R}^5$$

$$\dim\big(\boldsymbol{R}(A)\big) + \dim\big(\boldsymbol{N}(A)\big) = 5$$

$$\dim\big(\boldsymbol{C}(A)\big) = 2, \qquad \boldsymbol{C}(A) \subseteq \mathbb{R}^3$$

$$\dim\left(\boldsymbol{N}\big(A^T\big)\right) = 1, \qquad \boldsymbol{N}\big(A^T\big) \subseteq \mathbb{R}^3$$

$$\dim\big(\boldsymbol{C}(A)\big) + \dim\left(\boldsymbol{N}\big(A^T\big)\right) = 3$$